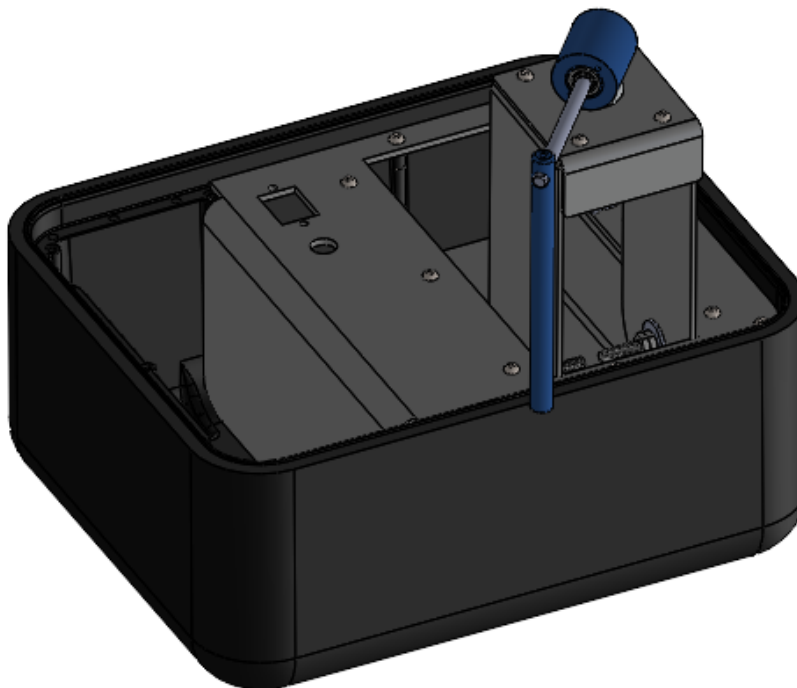


Open-Source Teaching Platform for Control Systems: Design Report

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1 Executive Summary

This project set out to design a low-cost, open-source replacement for the Quanser Qube-Servo 2, a \$5,000 device used in control systems classes. The target was a fully functional prototype delivered by May 2026 at under \$1,000 per unit, capable of supporting the same lab experiments, including angular position control with an inertia disc and inverted pendulum balancing, while being self-contained, portable, and compatible with MATLAB/Simulink.

The design integrates all components into a Nanuk 905 protective case with built-in storage for attachments, cables, and the power supply. The system uses an Arduino microcontroller, a motor driver, and a custom PCB, with two interchangeable magnetic attachments, an inertia disc and a pendulum, mirroring the Qube-Servo 2's functionality. The projected per-unit production cost of \$763.68 comes in well under the \$1,000 target.

The device met most implementation requirements. Data transmission, MATLAB/Simulink compatibility, third-party manufacturability, self-contained storage, and transport protection all passed evaluation. Open-source documentation, including CAD files, a bill of materials, and assembly instructions, has been prepared and made publicly available.

However, the device did not fully replicate the closed-loop control performance of the Qube-Servo 2. Arduino's data transmission rate is approximately an order of magnitude slower than the proprietary controller used in the Qube-Servo 2, causing significant control loop delays and excessive overshoot during inertia disc testing. As a result, the pendulum balancing test was not attempted.

The primary recommended next step is to replace the Arduino with a faster microcontroller, retest the system using the same control experiments, and proceed to a small production run once performance requirements are confirmed to be met.

2 Problem Definition

2.1 Problem Scope

The purpose of this project is to design a device for students taking control systems to gain hands-on experience with the implementation of control systems, such as PID circuits, to run simple tasks on a real mechanical device. This device will replace the Quanser Qube-Servo 2, which is currently utilized by the instructor to accomplish these goals. The device must function similarly to the Qube-Servo 2, and mechanical components such as the DC motor must maintain similar specifications to ensure the device can be used to complete the projects that are completed as part of the control systems class. The team was tasked with developing a working prototype of this device, testing this device to ensure the performance matches the expectations of the project sponsors, and completing the preparation work that will be necessary for the sponsors to order all the components of the device from third-party manufacturing companies. This device must be open source and have the ability to be updated so that it will not go obsolete as the software it depends on goes obsolete. A fully functioning final prototype of this device is required to be delivered by May 1, 2026, and all necessary information necessary to reproduce the device, such as CAD files, a bill of materials, assembly steps, and clearly documented code, must be delivered to the sponsors by May 11, 2026. The design team was not responsible for a full production run of the device or acquiring certifications for the device. The design of this project is focused on replicating the functionality of the Qube-Servo 2 and only making changes necessary to meet the design requirements, not adding additional attachments to the device or expanding the number of projects that can be completed by the device.

2.2 Technical Review

As the purpose of the project is to design a device to replace the Qube-Servo 2, most of the reviewed material relates directly to that system. The documentation and user instructions for the Qube-Servo 2 were used to identify key specifications for major components such as the motor and encoders. This includes values like the motor torque constant and rotor inertia, as well as encoder line count and resolution. These values, along with other relevant parameters, are listed in a specifications table in the Appendix and were used as a baseline for comparison. During the review of the device, it was found that the Qube-Servo 2 uses a proprietary microcontroller. Because of this, the original control code could not be accessed or reviewed, which limits direct understanding of how the system operates at

the software level. The device also uses a separate motor controller to send commands from the microcontroller to the motor, along with an additional circuit board to reduce internal wiring and manage connections between components. When the pendulum attachment was disassembled, it was found to consist of three custom-machined parts, two bearings, an optical encoder, and standard fasteners. The inertia disk attachment follows a similar structure. Both attachments connect to the motor using magnets, which allows them to be attached and removed without tools. The overall system is made up of several separate components. These include the main cube, which houses the motor and control hardware, the pendulum and inertia disk attachments, a USB cable for communication with a computer, and an external power supply. These components are not integrated into a single enclosure and are stored separately when not in use.

2.3 Design Requirements

Based on the needs and goals defined by the project sponsors, the following design requirements were selected to guide development. The device must replicate the core functions of the Qube-Servo 2 so that it can be used to complete the same control systems experiments. This includes moving the inertia disk between set angles while accounting for overshoot caused by rotational inertia, as well as balancing the pendulum in the upright position. To support this, the motor and encoder must have similar performance to those used in the Qube-Servo 2, and the system must be able to collect and transfer data at speeds that match or exceed the existing device. All aspects of the device must be open source. This includes CAD files, electrical layouts, code, and assembly instructions. These materials must be publicly available in a GitHub repository so that the device can be reproduced without relying on proprietary information. Cost is a major requirement. The Qube-Servo 2 costs about \$5000 per unit, and the target for this project is about \$1000 per unit. This requires careful selection of components and manufacturing methods while still meeting performance needs. The device must be self-contained, with all parts stored together in a way that prevents loss or damage. It should not rely on loose storage such as a cardboard box. The system must also be protected during transport and be easy for students to carry between locations. The system must be compatible with MATLAB and Simulink, and the software must be reprogrammable so it can be updated over time as these tools change. It should also be easy to perform maintenance, with components that can be accessed and replaced without difficulty. All parts must be able to be manufactured by third-party vendors, with final assembly completed by the end user using the provided documentation.

3 Design Description

3.1 Overview

The design maintains the core functionality and accessory compatibility of the Qube-2 while reconfiguring the system into a portable, self-contained unit. Instead of a cube intended for stationary tabletop use, the device is integrated into a protective case that supports transport, storage, and operation within a single structure. The case incorporates features to support use and organization, including a deployable mounting platform and internal storage for attachments and cables. The internal structure is fixed within the case to maintain alignment and stability during operation.

The device is used by students in a control systems course to implement and test real-time control algorithms. A laptop is connected to the device, and Simulink is used to develop programs that control the motor based on project objectives. These programs process sensor data that tracks motor position and generate corresponding control responses in real time.

The device supports two primary attachments. The first is an inertia disc, which is used to study angular position control while accounting for overshoot resulting from the disc's angular momentum. The second is a pendulum attachment that includes an additional sensor to track its angular position. This configuration is used for control problems where motor motion is adjusted to maintain the pendulum in an upright (inverted) position.

3.2 Detailed Description

The final device is shown in Figure 1. In this figure, the device is shown in its operation position with all subassemblies installed.



Figure 1 Image showing the full final design

The device is contained in a Nanuk 905 case, which can be closed and latched for storage and transportation. The lid of the case contains a storage compartment to store the device attachments when not in use. The cavity on the left-hand side of the device is used to store the power supply and data cable. The front panel includes the power and data ports, and also the motor housing.

A close-up view of the attachment storage compartment is shown in Figure 2.



Figure 2 Image showing the attachment storage compartment

The storage compartment contains two cavities, one for the inertia disc attachment and one for the pendulum attachment. While being stored, the attachments are fixed in place using magnets. The compartment is made up of two layers of hardboard with a layer of foam in between. The top layer of hardboard is mounted to the case, and the foam and the other hardboard layer are attached using glue.

A view of the device in its storage state from above is shown in Figure 3.



Figure 3 Device in its storage state

As can be seen in this image, the motor housing is capable of folding down into a trough, which allows the lid to close for storage. To the left of this is a power port and a USB port, where the student would connect their device to their laptop. The cavity on the left, where the USB cable and power supply are stored, is divided from the rest of the internal section of the device with a section of the top panel that is bent down to create this division. This panel is made of anodized aluminum and is screwed to the Nanuk case.

Figure 4 depicts the electrical compartment, located behind the divider that forms the storage cavity.

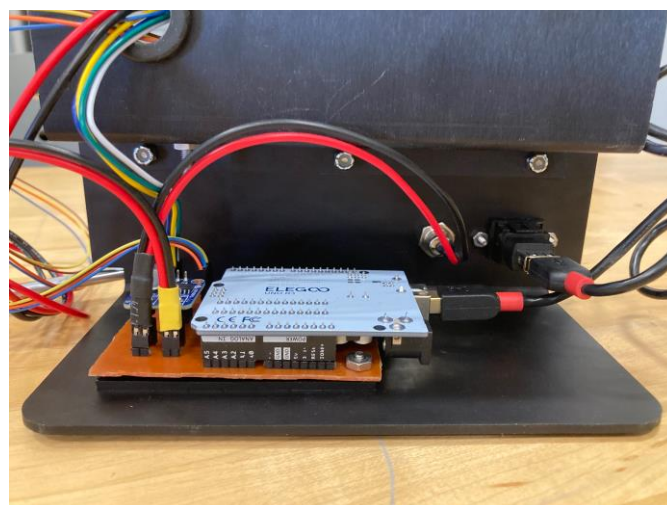


Figure 4 Electrical compartment of the device

The electrical compartment contains an Arduino, which is used as the microprocessor to control the device. A motor driver is also housed in this compartment and controls the voltage applied to the motor based on an input signal. Both the Arduino and the motor driver are connected to the printed circuit board (PCB). The PCB acts as a junction between the motor and encoder sensors and the motor and consolidates the wiring of the circuit. The Arduino and motor are both directly plugged into the PCB, and the PCB is mounted to the backside of the storage cavity divider with an acrylic plate on the back to insulate the signals in the PCB from the aluminum panel.

The motor housing is shown in the two images displayed in Figure 5.

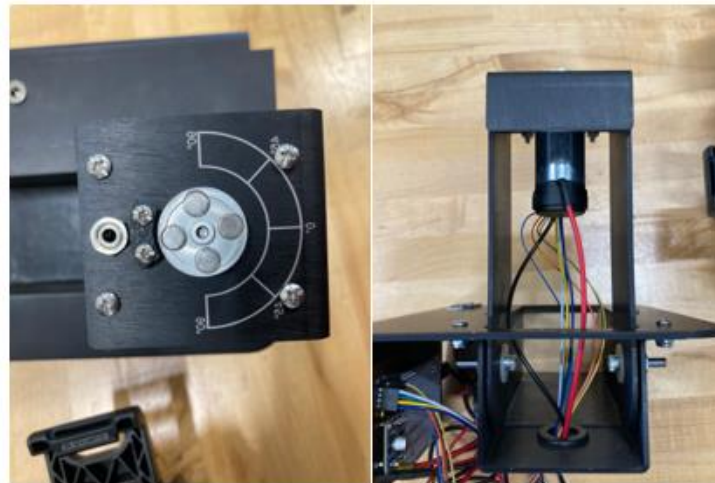


Figure 5 Top view of the motor housing (left) and side view of the motor housing (right)

In the image on the left, the top of the motor housing, which contains a motor mounting plate that has a built-in handle, can be seen in the right image. The mounting plate has a protractor engraved on it to help demonstrate the control the device has over the inertia disc attachment. The motor is mounted to the bottom of this plate. An attachment interface component is attached to the end of the motor shaft, which utilizes magnets to affix the attachments. There is a 3.5 mm jack port on the plate, which allows the encoder in the pendulum attachment to be connected to the device while in use. The small semicircular component next to the motor is attached to the motor plate to protect the pendulum from swinging too far and damaging the wiring. Below the motor plate are the two motor housing legs, which are used to elevate the motor to allow the pendulum to swing freely while in use. The motor legs are then attached to the trough within the main device using a combination of nylon washers, nuts, and partially threaded bolts, which work as a friction-resistant hinge, keeping the motor housing in place while in the vertical and horizontal positions.

The two device attachments are shown in Figure 6.

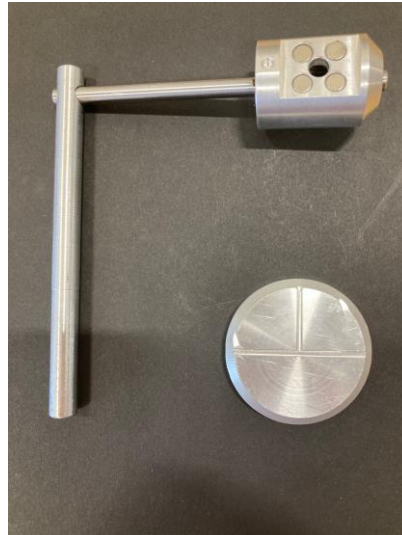


Figure 6 Image of the two device attachments

On the bottom right is the inertia disc. This disc is solid aluminum and is attached to the device to add inertia to the motor. On the front, there are lines cut into the material that are used to help the observer track the position of the motor and see physically what the device is doing. The disc has magnets inserted into holes on the back, which allow the attachment to be mounted to the mounting interface on the motor. The pendulum consists of 3 major components and some hardware. The pendulum cylinder is the component that attaches to the motor via the attachment interface with the magnets that can be seen in the image. The pendulum axle runs through bearings that are press-fit into the ends of the cylinder and are secured using retaining rings. The pendulum encoder is then mounted to one end of the pendulum cylinder and tracks the position of the pendulum axle. This encoder is connected to a 3.5 mm plug that plugs into the jack on the motor plate when in use. This is not shown in the image, as it was determined that it was not worth wasting an encoder for a pendulum attachment that would not be used. On the far end of the axle is the pendulum arm, which is fixed to the axle using a set screw and creates a moment that causes the pendulum to rotate when out of perfect alignment.

3.3 Use

The device is used in a control systems lab to implement and test real-time feedback control algorithms. A laptop is connected to the system, and Simulink is used to write and deploy control programs that interface with the motor and encoders. The system runs in a closed loop, with encoder data fed back to the controller so motor input can be updated in real time.

The first application uses the inertia disc attachment for angular position control. The motor moves the disc between set positions while accounting for inertia and limiting overshoot. This is used to tune PID controllers and observe closed-loop response.

The second application uses the pendulum attachment. The controller moves the pendulum from a downward position to an upright position and maintains it there. This inverted pendulum setup is used to study system stability and disturbance response.

Across both cases, the device is used to connect control theory developed in simulation with a physical system.

4 Design Evaluation

4.1 Overview

This section outlines how the final design was evaluated against the requirements defined in the problem definition section to ensure that the final device meets the sponsor's expectations for the project. The key requirements and corresponding evaluation methods are summarized in Table 4.1.

Table 4.1 Design Requirements, Test Methods, and Targets

Requirements	Target Value	Testing Method
Qube Function Replication	Pass	Complete Qube Servo 2 Tasks
Motor and Encoder Performance	Comparable	Qube response comparison
Accurate Data Transmission	Pass	Send and receive data
Cost Target	< \$1,000.00	Hand Calculation
Self-Contained Storage	Pass	User Testing
Transport Protection	Pass	User Testing
Portability	Pass	User Testing
MATLAB/Simulink Compatibility	Pass	MATLAB/Simulink Connection
Software Updatability	Pass	Code Modification
Third-Party Manufacturability	Pass	Procurement Validation

4.2 Evaluation of Design Criteria

4.2.1 Qube Function Replication

Purpose

Evaluating whether the device replicates the functionality of the Qube-Servo 2 is critical to determining overall project success. The objective is to assess whether the system can support the same control experiments performed in the control systems course. If the device cannot reliably execute these experiments, it cannot serve as a replacement.

Test Methods

First, an angular position control project is tested using the inertia disc attachment. A Simulink program is developed to move the disc between specified angular positions (e.g., from 1 rad to -1 rad), with the expectation that the system will reach each position with limited overshoot and briefly stabilize before reversing direction.

Second, a vertical pendulum balancing project is defined. In this test, the pendulum attachment is used, and a Simulink program is designed to swing the pendulum from a downward position to an upright position and maintain it there through continuous motor adjustments. Successful completion requires stable balancing in the inverted position.

Results

The inertia disc test was performed, but was not successful. The microcontroller used in the design, an Arduino, operates with a data transmission rate approximately an order of magnitude slower than the proprietary controller used in the Qube-Servo 2. This limitation introduced a significant delay in the control loop.

As a result, the system was unable to respond to position feedback in a timely manner. During testing, when commanding motion between 1 rad and -1 rad using a PID controller, the motor consistently overshoot the target position by a large margin. In some cases, the motor position exceeded 100 rad before reversing direction. This behavior indicates that the controller was acting on outdated position data, leading to excessive acceleration and instability.

Due to the failure of the inertia disc test, the pendulum balancing test was not attempted. The inverted pendulum problem requires faster and more precise control than basic angular position regulation, and there was no indication that the system would perform adequately under those conditions given the observed limitations.

Overall, the device does not replicate the functional performance of the Qube-Servo 2 and does not meet the requirements for use in the intended control systems experiments.

4.2.2 Motor and Encoder Performance

Purpose

This test evaluates the motor and encoder response of the system to determine whether it can support control systems instruction. Accurate and repeatable motion tracking is required, and performance is expected to match the Qube-Servo 2 under equivalent conditions.

Test Methods

A Simulink-based motion profile is executed on both the device and the Qube-Servo 2. Each system is commanded to follow the same set of position changes over time, and encoder position is recorded and plotted for comparison against the input trajectory.

Performance is assessed by comparing how closely each system tracks the commanded motion in similar operating conditions. The device is expected to meet or exceed the tracking accuracy of the Qube-Servo 2 for this test to pass.

Results

This test is dependent on the system's ability to replicate the core functionality of the Qube-Servo 2 under closed-loop control conditions. As identified in Section 4.2.1, the device was not able to achieve equivalent performance in the inertia disc control task due to limitations in the control loop implementation.

Because this test evaluates the same motor and encoder response behavior under comparable operating conditions, the previously observed performance limitations directly carry over. The device is unable to maintain the required accuracy and stability in real-time motion tracking when compared to the Qube-Servo 2.

Based on this, the system does not meet the required performance equivalence for motor and encoder response.

4.2.3 Accurate Data Transmission

Purpose

Testing data transmission between Simulink and the Arduino is necessary to determine whether the control systems experiments can be completed successfully. If transmitted data is inaccurate or inconsistent, the inertia disc project will not reach correct target positions, and the pendulum system will not receive correct feedback for maintaining stability.

Test Methods

Data transmission is evaluated by comparing commanded and returned values within Simulink and the Arduino system. The following test cases are used:

1. **Bidirectional Simulink-Arduino communication (loopback test)**
Simulink sends numerical values to the Arduino, which are immediately returned to Simulink without interacting with hardware components.
This verifies that transmitted values remain unchanged through the communication process.
2. **Encoder value transmission to Simulink**
Encoder readings are transmitted from the Arduino to Simulink in real time.
This test evaluates whether position data is correctly represented in Simulink without loss or distortion.

3. **Motor command transmission (voltage control)**

Simulink sends voltage commands to the Arduino to control motor direction and magnitude. The received values are checked against the commanded inputs to confirm correct interpretation and execution.

4. **Closed-loop motor control with encoder feedback**

Voltage commands are sent to the motor while encoder values are returned to Simulink. This test evaluates whether transmitted command and feedback signals remain accurate when operating together in a full control loop.

Results

1. Pass
2. Pass
3. Pass
4. Pass

4.2.4 Cost Target

Cost Objective

Cost was one of the primary constraints for the project. The Quanser Qube-Servo 2 costs \$5000 per unit, and the target for this design was to keep the cost below \$1000 per unit. This requirement is critical because the value of this project must replicate the functionality of the Qube 2 and do so at a realistic cost for universities to adopt and reproduce the system at scale. A device that performs well but remains too expensive to replicate would not meet the intent of the project.

For this reason, success in this section is defined by whether the final design can be reproduced at or below the \$1000 target per unit. Prototype development costs are considered separately, as they include one-time expenses and inefficiencies that do not represent final production.

Methodology and Assumptions

The cost of the system was evaluated from the perspectives of total research and development cost for the prototype and projected per-unit cost assuming a production run of ten units. The R&D cost includes all expenses related to components, manufacturing, and design iterations, and reflects the true cost of development. The projected per-unit cost was calculated using a finalized bill of materials, supplier quotes, and known manufacturing costs. Batch production was assumed where applicable, particularly for attachment and housing components, reducing per-unit cost compared to prototype manufacturing. Labor and assembly were not included, and prototype-specific costs such as duplicate purchases and redesigned iterations were excluded.

Resulting Budget

The estimated per-unit cost for the final design is \$763.68, well below the \$1000 target, providing a margin of roughly \$236. This indicates the design meets cost requirements and is significantly more affordable than the Qube-Servo 2.

Total research and development cost is \$1739.75, compared to a running estimate of \$1187.00. The higher true cost is mainly due to manufacturing costs. Originally estimated around \$280.00, a figure closer to \$1000 is more realistic due to low-volume production and design iteration. Additional increases came from motor selection changes and higher electrical component costs. Despite this, the projected production cost remains within the target, as prototype inefficiencies such as one-off manufacturing and duplicate purchases would not apply to batch production. Overall, while development costs exceeded initial estimates, the final design meets the cost objective and remains economically viable.

4.2.5 Self-Contained Storage

Purpose

The Quanser Qube-Servo 2 is made up of five components that are all loose and currently stored together in a cardboard box. The device designed for this project solves this issue by being built in a

case with storage compartments to fit all of the loose components. If even one of these components is unable to be stored in the case, it would defeat the purpose of building the device within a case.

Test Methods

All the loose components of the device: the 2 attachments, the power supply, and the computer connection cable, will be placed in their respective storage locations. If the case can be closed without damaging any of the components, the device will pass the test.

Results

The first prototype of the device was tested by placing only the power supply and computer connection cable in the storage compartment, and it was determined that the compartment was too small to accommodate these components. This was increased in the second iteration, which had a large enough compartment to accommodate the cords. When the attachment storage compartment in the lid was added to the device, the attachments were placed in their respective locations, the cords were placed in their storage compartment, and the lid was closed. The lid was able to close completely, and the Nanuk case was successfully latched shut, indicating that the device passed the test.

4.2.6 Transport Protection

Purpose

The device contains many components that can be easily damaged if exposed to water, if they are bumped, or dropped, and so it was decided to integrate the device within a Nanuk 905 case to protect these delicate components from getting damaged. The extent to which this case will protect the internal components will not be tested, as destructive testing is not within the budget of this project. The Nanuk 905 case is, however, certified to protect contents from rain and drops as per ASTM D-4169 DC-18. The certification that this case has will be assumed to ensure the safety of the internal contents.

4.2.7 MATLAB/Simulink Compatibility

Purpose

Being able to connect the device to MATLAB/Simulink allows data to be transferred between the controlling script written by the student and the Arduino within the device. Compatibility with MATLAB/Simulink is critical to the function of the device, and the project will not be successful without it.

Test Methods

An Arduino code was created that uses serial transmission and a receiver, along with a MATLAB Systems Simulink block that can utilize serial communication to confirm that signals can be transmitted in both directions. The test will be successful if the values are sent from Simulink to Arduino and then returned, and the values received at the Arduino and returned to Simulink match the original value.

Results

The test values were sent, and the values received at the Arduino and received in Simulink matched the values that were sent, which means the device passed the test.

4.2.8 Software Updatability

Purpose

A known limitation of the Qube-2 system is that it only operates reliably with older versions of MATLAB and Simulink, creating compatibility issues for current classroom use. This project aims to ensure the

device functions on modern and recent software versions without requiring extensive configuration or troubleshooting. The goal is a straightforward setup process where the system can be used directly in a typical control system.

Test Methods

Initial testing involved running the MATLAB, Simulink, and Arduino code on the latest software version as well as several recent supported versions. When issues were encountered on the most recent release, testing was adjusted to evaluate compatibility on earlier versions instead. The modified test focused on verifying correct motor voltage control and encoder-based position tracking using Simulink across those versions.

Results

The system did not operate correctly on the latest MATLAB and Simulink release, requiring a revision of the test approach. On earlier supported versions, the device was able to successfully control motor voltage and accurately track position using encoder feedback. Behavior was consistent across the tested older versions, with no observed data or control issues. Based on these results, the system is considered compatible with prior supported software versions. It is also expected that future compatibility can be maintained with minor code or syntax updates as software versions change. Based on these results, the device passes the test.

4.2.9 Third-Party Manufacturability

Purpose

The sponsors of this project would like all the components of the device to be able to be manufactured by third-party companies, so that all parts can be ordered and then just assembled. This will also mean that any other individuals who are interested in replicating the device will be able to just order all the components and follow the assembly instructions to complete the device.

Test Methods

All components required to build the device will be ordered by third-party manufacturing companies, and if a suitable component is received for each of these components, the device will pass this test.

Results

In the first two prototypes, issues with the manufacturability were found and subsequently solved. All parts that have been received at this point have been confirmed to be manufacturable. This includes the main plate, the trough, the motor housing legs, and the motor plate.

4.2.10 Device Evaluation Summary

The device meets several key requirements, including accurate data transmission, MATLAB/Simulink compatibility, open-source documentation, cost targets, self-contained storage, transport protection, and software updatability on supported versions. These results show the system is usable, reproducible, and deployable in a control systems lab with minimal setup. The design also performs well in packaging and integration, with all components successfully stored and protected within the case.

The device does not achieve functional equivalence with the Qube-Servo 2 in closed-loop control performance. The inertia disc test failed due to limitations in the control loop, which affected motor and encoder performance under dynamic conditions. As a result, the inverted pendulum test was not completed. Ease of maintenance is assumed to pass based on planned instruction manual revisions and user testing. Overall, the device meets most implementation requirements but does not meet the performance requirements for full replacement of the Qube-Servo 2.

4.3 Conclusions

The final design addresses many of the implementation and usability goals established at the beginning of the project. The device was redesigned into a self-contained portable device that integrates storage, protection, and operation into a single system while using similar attachments as the Qube-2. The system met several major project requirements, including accurate data transmission, MATLAB/Simulink compatibility, open-source documentation, manufacturability through third-party vendors, and meets the overall cost target. The projected production cost of approximately \$764 per unit is significantly lower than the cost of the Qube-Servo 2, demonstrating that a much more affordable alternative can be developed while keeping and improving many of the same functional and educational features.

The physical design of the device was effective for the presentation and mechanical function of the device. The integrated Nanuk case successfully stores and protects all system components, improving portability and organization compared to the existing Qube-Servo 2 design. The modular assembly process and open-source hardware and software support long-term maintainability and reproducibility for future users or institutions.

Despite these successes, the project did not fully achieve the primary performance objective of replicating the operational functionality of the Qube-Servo 2. Testing showed that the Arduino-based control system was unable to process and transmit feedback data at the speed required for stable real-time control. This resulted in significant delays between input and output during inertia disc testing, preventing the successful completion of the intended control experiments and making pendulum balancing impossible with the current hardware configuration. Because of this limitation, the device cannot yet serve as a complete replacement for the Qube-Servo 2 in its current form.

Overall, the project demonstrated that a low-cost, open-source, and manufacturable control systems teaching platform is feasible. The results also identified the primary technical limitation preventing full success, providing a clear direction for future development. Replacing the current microcontroller with a faster platform and retesting the system would likely allow the design to achieve the required closed-loop performance needed to operate the pendulum, while retaining the strengths of the existing mechanical and electrical architecture.

4.4 Next Steps

The next step in the project is to address the primary limitation identified in testing, which is the performance of the Arduino microcontroller. A replacement microcontroller with higher data handling capability needs to be selected and integrated into the existing system architecture. Once integrated, the full system should be retested using the same control experiments that were previously used, including the inertia disc and pendulum tests, to confirm that the updated control loop meets the required performance and resolves the instability observed in this iteration.

After the system successfully passes these tests, the remaining steps can proceed as planned. This includes cataloging all existing components from the prototype phase to ensure reusable parts are not wasted. A small production run can then be assembled using ordered components, with assembly completed in batches by sophomore or junior mechanical or electrical engineering students following the finalized instruction manual. Each completed unit should undergo functional testing before deployment, and a prototype unit should be provided to the course instructor for preparation and integration into the control systems curriculum.

5 Appendix

5.1 Quanser Qube-Servo 2 Specifications Table

Symbol	Description	Value
DC Motor		
V_{nom}	Nominal input voltage	18.0 V
τ_{nom}	Nominal torque	22.0 mN-m
ω_{nom}	Nominal speed	3050 RPM
I_{nom}	Nominal current	0.540 A
R_m	Terminal resistance	8.4 Ω
k_t	Torque constant	0.042 N-m/A
k_m	Motor back-emf constant	0.042 V/(rad/s)
J_m	Rotor Inertia	4.0×10^{-6} kg-m ²
L_m	Rotor inductance	1.16 mH
m_h	Module attachment hub mass	0.0106 kg
r_h	Module attachment hub radius	0.0111 m
J_h	Module attachment moment of Inertia	0.6×10^{-6} kg-m ²
Inertia Disc Module		
m_d	Disc mass	0.053 kg
r_d	Disc radius	0.0248 m
Rotary Pendulum Module		
m_r	Rotary arm mass	0.095 kg
L_r	Rotary arm length (pivot to end of metal rod)	0.085 m
m_p	Pendulum link mass	0.024 kg
L_p	Pendulum link length	0.129 m
Motor and Pendulum Encoders		
	Encoder line count	512 lines/rev
	Encoder line count in quadrature	2048 lines/rev
	Encoder resolution (in quadrature, deg)	0.176 deg/count
	Encoder resolution (in quadrature, rad)	0.00307 rad/count
Amplifier		
	Amplifier type	PWM
	Peak Current	2 A
	Continuous Current	0.5 A
	Output voltage range (recommended)	± 10 V
	Output voltage range (maximum)	± 15 V

Table 2.2: QUBE-Servo 2 System Parameters